LEVEL 1: EASY (Powers 2 to 3)

Exercise 1

$$(x+3)^2$$

Answer 1

Pascal Row 2: $1, 2, 1$

$$1(x)^2(3)^0 + 2(x)^1(3)^1 + 1(x)^0(3)^2$$

$$= x^2 + 2(x)(3) + 9$$

$$\mathbf{= x^2 + 6x + 9}$$

Exercise 2

$$(2a-1)^3$$

Answer 2

Pascal Row 3: $1, 3, 3, 1$

$$\begin{aligned} &= 1(2a)^3 + 3(2a)^2(-1) \\ &\quad + 3(2a)^1(-1)^2 + 1(-1)^3 \\ &= 8a^3 + 3(4a^2)(-1) + 3(2a)(1) - 1 \end{aligned}$$

$$\mathbf{= 8a^3 - 12a^2 + 6a - 1}$$

Exercise 3

$$(y+2)^4$$

Answer 3

Pascal Row 4: $1, 4, 6, 4, 1$

$$\begin{aligned} &= 1(y)^4 + 4(y)^3(2) + 6(y)^2(2)^2 \\ &\quad + 4(y)^1(2)^3 + 1(2)^4 \\ &= y^4 + 8y^3 + 6(y^2)(4) + 4(y)(8) + 16 \end{aligned}$$

$$\mathbf{= y^4 + 8y^3 + 24y^2 + 32y + 16}$$

LEVEL 2: INTERMEDIATE (Powers with Exponents & Higher Degrees)

Exercise 4

$$(x^2 - 3)^3$$

Answer 4

Pascal Row 3: $1, 3, 3, 1$

$$\begin{aligned} &= 1(x^2)^3 + 3(x^2)^2(-3)^1 \\ &\quad + 3(x^2)^1(-3)^2 + 1(-3)^3 \\ &= x^6 + 3(x^4)(-3) + 3(x^2)(9) - 27 \end{aligned}$$

$$\mathbf{= x^6 - 9x^4 + 27x^2 - 27}$$

Exercise 5

$$(3x + y^2)^4$$

Answer 5

Pascal Row 4: $1, 4, 6, 4, 1$

$$\begin{aligned} &= 1(3x)^4 + 4(3x)^3(y^2)^1 \\ &\quad + 6(3x)^2(y^2)^2 + 4(3x)^1(y^2)^3 + 1(y^2)^4 \\ &= 81x^4 + 4(27x^3)(y^2) \\ &\quad + 6(9x^2)(y^4) + 12xy^6 + y^8 \end{aligned}$$

$$\mathbf{= 81x^4 + 108x^3y^2 + 54x^2y^4 + 12xy^6 + y^8}$$

Exercise 6

$$(a - 2b)^5$$

Answer 6

Pascal Row 5: $1, 5, 10, 10, 5, 1$

$$\begin{aligned} &= 1(a)^5 + 5(a)^4(-2b) + 10(a)^3(-2b)^2 \\ &\quad + 10(a)^2(-2b)^3 + 5(a)(-2b)^4 + 1(-2b)^5 \\ &= a^5 - 10a^4b + 10a^3(4b^2) \\ &\quad - 10a^2(8b^3) + 5a(16b^4) - 32b^5 \end{aligned}$$

$$\mathbf{= a^5 - 10a^4b + 40a^3b^2 - 80a^2b^3 + 80ab^4 - 32b^5}$$

Exercise 7

$$(2x^2 + 3y)^3$$

Answer 7

Pascal Row 3: $1, 3, 3, 1$

$$\begin{aligned} &= 1(2x^2)^3 + 3(2x^2)^2(3y)^1 \\ &\quad + 3(2x^2)^1(3y)^2 + 1(3y)^3 \\ &= 8x^6 + 3(4x^4)(3y) + 3(2x^2)(9y^2) + 27y^3 \end{aligned}$$

$$= \mathbf{8x^6 + 36x^4y + 54x^2y^2 + 27y^3}$$

LEVEL 3: HARD (Complex Exponents & Rational Expressions)

Exercise 8

$$(x^3 - 2y^2)^4$$

Answer 8

Pascal Row 4: $1, 4, 6, 4, 1$

$$\begin{aligned} &= 1(x^3)^4 + 4(x^3)^3(-2y^2)^1 \\ &\quad + 6(x^3)^2(-2y^2)^2 \\ &\quad + 4(x^3)^1(-2y^2)^3 + 1(-2y^2)^4 \\ &= x^{12} + 4(x^9)(-2y^2) + 6(x^6)(4y^4) \\ &\quad + 4(x^3)(-8y^6) + 16y^8 \end{aligned}$$

$$= \mathbf{x^{12} - 8x^9y^2 + 24x^6y^4 - 32x^3y^6 + 16y^8}$$

Exercise 9

$$\left(2x^2 - \frac{1}{x}\right)^5$$

Answer 9

Pascal Row 5: $1, 5, 10, 10, 5, 1$

$$\begin{aligned}
&= 1(2x^2)^5 + 5(2x^2)^4\left(-\frac{1}{x}\right)^1 \\
&\quad + 10(2x^2)^3\left(-\frac{1}{x}\right)^2 + 10(2x^2)^2\left(-\frac{1}{x}\right)^3 \\
&\quad + 5(2x^2)^1\left(-\frac{1}{x}\right)^4 + 1\left(-\frac{1}{x}\right)^5 \\
&= 32x^{10} - 5(16x^8)\left(\frac{1}{x}\right) + 10(8x^6)\left(\frac{1}{x^2}\right) \\
&\quad - 10(4x^4)\left(\frac{1}{x^3}\right) + 5(2x^2)\left(\frac{1}{x^4}\right) - \frac{1}{x^5}
\end{aligned}$$

$$= \mathbf{32x^{10} - 80x^7 + 80x^4 - 40x + \frac{10}{x^2} - \frac{1}{x^5}}$$

Exercise 10

$$(3a^3b - 2c^2)^4$$

Answer 10

Pascal Row 4: $1, 4, 6, 4, 1$

$$\begin{aligned}
&= 1(3a^3b)^4 + 4(3a^3b)^3(-2c^2)^1 \\
&\quad + 6(3a^3b)^2(-2c^2)^2 \\
&\quad + 4(3a^3b)^1(-2c^2)^3 + 1(-2c^2)^4 \\
&= 81a^{12}b^4 + 4(27a^9b^3)(-2c^2) \\
&\quad + 6(9a^6b^2)(4c^4) + 4(3a^3b)(-8c^6) + 16c^8
\end{aligned}$$

$$= \mathbf{81a^{12}b^4 - 216a^9b^3c^2 + 216a^6b^2c^4 - 96a^3bc^6 + 16c^8}$$

LEVEL 1: EASY (Basic Monomials and Binomials)

Exercise 1

$$(3x)(5x^2)$$

Answer 1

Multiply coefficients and add exponents:

$$(3 \cdot 5) \cdot (x^1 \cdot x^2)$$

$$= 15x^{1+2}$$

$$\boxed{= \mathbf{15x^3}}$$

Exercise 2

$$(2x)(4x + 7)$$

Answer 2

Apply distributive property:

$$(2x)(4x) + (2x)(7)$$

$$= 8x^2 + 14x$$

$$\boxed{= \mathbf{8x^2 + 14x}}$$

Exercise 3

$$(x + 4)(x + 6)$$

Answer 3

Apply FOIL:

1. **First:** $x \cdot x = x^2$
2. **Outer:** $x \cdot 6 = 6x$
3. **Inner:** $4 \cdot x = 4x$
4. **Last:** $4 \cdot 6 = 24$

$$\boxed{= \mathbf{x^2 + 10x + 24}}$$

LEVEL 2: INTERMEDIATE (Binomial $\times$ Binomial)

Exercise 4

$$(2x+3)(5x+4)$$

Answer 4

FOIL steps:

1. **First:** $(2x)(5x) = 10x^2$
2. **Outer:** $(2x)(4) = 8x$
3. **Inner:** $(3)(5x) = 15x$
4. **Last:** $(3)(4) = 12$

$$= \mathbf{10x^2 + 23x + 12}$$

Exercise 5

$$(3a^2+2)(4a^3+5)$$

Answer 5

FOIL steps:

1. **First:** $(3a^2)(4a^3) = 12a^5$
2. **Outer:** $(3a^2)(5) = 15a^2$
3. **Inner:** $(2)(4a^3) = 8a^3$
4. **Last:** $(2)(5) = 10$

$$= \mathbf{12a^5 + 8a^3 + 15a^2 + 10}$$

Exercise 6

$$(4x+3y)(2x+5y)$$

Answer 6

FOIL steps:

1. **First:** $(4x)(2x) = 8x^2$
2. **Outer:** $(4x)(5y) = 20xy$
3. **Inner:** $(3y)(2x) = 6xy$
4. **Last:** $(3y)(5y) = 15y^2$

$$= \mathbf{8x^2 + 26xy + 15y^2}$$

Exercise 7

$$(5m^2n+2)(3m+4n^3)$$

Answer 7

FOIL steps:

1. **First:** $(5m^2n)(3m) = 15m^3n$
2. **Outer:** $(5m^2n)(4n^3) = 20m^2n^4$
3. **Inner:** $(2)(3m) = 6m$
4. **Last:** $(2)(4n^3) = 8n^3$

$$= \mathbf{15m^3n + 20m^2n^4 + 6m + 8n^3}$$

LEVEL 3: HARD (Trinomials and Higher Degree)

Exercise 8

$$(2x+3)(x^2+4x+5)$$

Answer 8

Distribute each term:

1. $2x(x^2 + 4x + 5) = 2x^3 + 8x^2 + 10x$
2. $3(x^2 + 4x + 5) = 3x^2 + 12x + 15$

Combine like terms:

$$= \mathbf{2x^3 + 11x^2 + 22x + 15}$$

Exercise 9

$$(3a^2 + 2b)(a^2 + 4ab + 3b^2)$$

Answer 9

Distribute each term:

1. $3a^2(a^2 + 4ab + 3b^2) = 3a^4 + 12a^3b + 9a^2b^2$
2. $2b(a^2+4ab+3b^2) = 2a^2b+8ab^2+6b^3$

Group terms:

$$= \mathbf{3a^4 + 12a^3b + 9a^2b^2 + 2a^2b + 8ab^2 + 6b^3}$$

Exercise 10

$$(2x^2 + 3x + 1)(x^2 + 5x + 4)$$

Answer 10

Distribute each term:

1. $2x^2(x^2 + 5x + 4) = 2x^4 + 10x^3 + 8x^2$
2. $3x(x^2 + 5x + 4) = 3x^3 + 15x^2 + 12x$
3. $1(x^2 + 5x + 4) = x^2 + 5x + 4$

$$= \mathbf{2x^4 + 13x^3 + 24x^2 + 17x + 4}$$